Accumulation Functions and the Fundamental Theorem

- Each function below accumulates a measured *rate*. The units of the accumulated amount are the rate's units times the units of the variable being integrated.
- You do not need to find an antiderivative to differentiate an accumulation function — that is the point of the Fundamental Theorem.
- Give exact values (radicals, e) unless a decimal is requested.

1. Water enters a tank at $r(t) = 3t^2 + 4$ litres per minute. The volume added during the first t minutes is $V(t) = \int_0^t \left(3u^2 + 4\right)\, du$ litres. Find $V'(5)$, in litres per minute.

Answer: ________________

2. Using the same tank and the same V, find the volume of water added during the first 5 minutes.

Answer: ________________ L

3. A culture is counted by $P(x) = 200 + \int_0^x \frac{12}{1+t}\,dt$ thousand cells, where x is in days and $0 \le x \le 10$.

(a) Find $P'(3)$, in thousand cells per day.

Answer: ________________

(b) Explain why the culture never shrinks on this interval.

Answer: ________________

4. A sensor accumulates charge $Q(x) = \int_1^{x^2} \sqrt{1+t^3}\,dt$ millicoulombs, where x is in seconds.

(a) Find the exact value of $Q'(2)$.

Answer: ________________

(b) Give $Q'(2)$ to two decimal places.

Answer: ________________

5. A cooling tower still holds $C(x) = \int_x^{12} (0.5t + 2)\, dt$ litres of coolant after x hours of operation, for $0 \leq x \leq 12$.

(a) Find $C'(4)$, in litres per hour.

Answer: ________________

(b) Explain why $C'(x)$ is negative even though the integrand is positive.

Answer: ________________

6. Water flows into and out of a holding pond at the net rate $r(t) = 6 - t$ cubic metres per minute for $0 \leq t \leq 10$ (a negative rate means the pond is losing water). The change in volume over the first x minutes is $V(x) = \int_0^x (6 - t)\, dt$ cubic metres.

(a) Find $V'(8)$.

Answer: ________________

(b) Find the value of x at which V is largest.

Answer: ________________

(c) Justify that this value really gives the largest V on $[0, 10]$.

Answer: ________________

7. For the same pond and the same V:

(a) Find $V(6)$.

Answer: ________________ m^3

(b) Find $V(10)$.

Answer: ________________ m^3

8. A detector's reading is $F(x) = \int_{2x}^{x^2} e^{-t^2}\,dt$, with x in seconds.

(a) Find the exact value of $F'(1)$.

Answer: ________________

(b) Give $F'(1)$ to three decimal places.

Answer: ________________

(c) Explain why $F'(x)$ is not simply $e^{-x^4} - e^{-4x^2}$.

Answer: ________________